OIL SEEDS

COCONUT (*Cocos nucifera*)

Coconut requires an equatorial climate with high humidity. The ideal mean annual temperature is 27°C with 5-7° diurnal variation. The palm does not withstand prolonged spells of extreme variations. A well-distributed rainfall of 1300-2300 mm per annum is preferred.

Coconut is grown in different soil types such as laterite, coastal sandy, alluvial and also in reclaimed soils of the marshy lowlands. It tolerates salinity and a wide range of pH (from 5.0-8.0).

The cultural practices have to be adopted to suit the varying climatic and soil conditions.

Selection of site

Select sites with deep (not less than 1.5 m depth) well drained soil. Avoid shallow soils with underlying hard rock, low-lying areas subject to water stagnation and heavy clayey soils.

Cultivars

1. West Coast Tall (WCT)
2. Lakshadweep Ordinary (Chandrakalpa)
3. Philippines Ordinary (Kerachandra)
4. Andaman Ordinary
5. Java
6. Cochin China
7. Kappadam
8. Komadan
9. Kerasagara
10. Kalparaksha
11. Kalpadhenu
12. Kalpaprathibha
13. Kalpamithra

Hybrids

1. Lakshaganga (Lakshadweep Ordinary x Gangabondam)
2. Anandaganga (Andaman Ordinary x Gangabondam)
3. Keraganga (West Coast Tall x Gangabondam)
4. Kerasankara (West Coast Tall x Chowghat Orange Dwarf)
5. Chandrasankara (Chowghat Orange Dwarf x West Coast Tall)
6. Kerasree (West Coast Tall x Malayan Yellow Dwarf)
7. Kerasoubaghya (WCT x SSA)
8. Chowghat Green Dwarf x West Coast Tall
9. Chandralaksha (Lakshadweep Ordinary x Chowghat Orange Dwarf)

Tender nut variety: Chowghat Orange Dwarf

Keramadhura : A superior coconut suitable for tender nut and copra, with more quantity of nut water (287 ml) with excellent quality. Yield -119 nuts/palm; copra yield (196 g/nut).

Note:

1. Hybrids Anandaganga, Keraganga and Kerasankara are recommended for general cultivation both under rainfed and irrigated conditions.
2. Other hybrids especially Chandrasankara are recommended for ideal situations and where good management practices are adopted.
3. Since the performance of Chandrasankara is markedly superior to that of WCT in root (wilt) affected areas, cultivation of Chandrasankara and Kalparaksha are preferred in such areas.

4. Chandralaksha, Lakshaganga, Chandra-kalpa and Kalpadhenu are recommended for cultivation under drought prone areas.
5. Kalparaksha is recommended for tender coconut. Kalpaprathibha is a dual purpose variety (copra and tender coconut).
6. Kalpamithra is recommended for rainfed conditions.

Selection of mother palms

Select mother palms having the following characters:

1. Regular bearing habit and yielding not less than 80 nuts / annum.
2. Age 20 years or more (5 years after reaching full bearing capacity). If the mother palms are the progeny of elite planting material and gives consistently higher yields for a period of not less than 6 years, seed nuts can be collected from such palms. There is no need for insisting 20 years as minimum age for mother palms in such conditions.
3. More than 30 fully opened leaves with short strong petioles and wide leaf base firmly attached to the stem.
4. Bearing at least 12 bunches of nuts with strong bunch stalks.
5. Bearing nuts of medium size and oblong shape.
6. Husked nuts should weigh not less than 600 g.
7. Mean copra content of 150 g per nut or more.

Avoid palms which (i) have long, thin and pendulous inflorescence stalks (ii) produce long, narrow, small sized or barren nuts (iii) show shedding of immature nuts in large numbers and (iv) are grown under favourable environmental conditions.

Collection and storage of seed nuts

Collect mature nuts (above 11 month old) during the period from December to May. Lowering of bunches by means of ropes may be done when the palms are tall and ground is hard. Discard nuts, which show improper development or other undesirable features. Store seeds in shade for a minimum period of 60 days prior to sowing in nursery. For storing, arrange the seed nuts with the stalk-end up over an 8 cm layer of sand in a shed and cover with sand to prevent drying of nut water. Up to five layers of nuts can be arranged one over the other. The nuts can also be stored in plots, provided the soil is sandy and the ground is sufficiently shaded. In the case of nuts harvested in May, heap them in partial shade, till husk is well dried and then sow them in the nursery.

Selection and preparation of site for nursery

Nursery sites should be well drained with light textured soil and with adequate but not too much shade. In open areas, provide shade during summer. Prepare beds of 1.5 m width and of convenient length with 75 cm space between beds. In areas where drainage is poor, prepare raised beds.

Before sowing, examine seed nuts and discard those without nut water and rotten kernels. Sow the nuts in the nursery after commencement of southwest monsoon during May-June.

Spacing of nuts

Plant the seed nuts at a spacing of 30 cm (between rows) x 30 cm (between nuts) with four or five rows per bed.

Method of planting seed nuts

Plant the seed nuts in the beds in trenches 25-30 cm deep and cover with soil so that top

portion of husk alone is visible. The nuts may be planted either horizontally with the widest of the segments at the top or vertically with stalk-end up. Vertical planting is preferable on account of convenience in transporting and lesser risk of seedling injury.

Care and management of nursery

Provide protective fencing to the nursery if it is located in open area. If the soil is sandy, provide mulching immediately after the cessation of monsoon rain. Irrigate the nursery once in two days during summer months. Keep the nursery beds free of weeds by periodic weeding. If termite is noticed, remove soil in the affected area upto a depth of about 15 cm and dust soil and nuts with carbaryl or chlorpyrifos. Repeat if attack persists. Periodically spray the plants with 1per cent bordeaux mixture or any other copper fungicide to prevent fungal infection.

Selection of seedlings

Remove seed nuts, which do not germinate within 6 months after sowing as well as those with dead sprouts. Select only good quality seedlings (9-12 months old) by a rigorous selection based on the following characteristics.

1. Early germination, rapid growth and seedling vigour.
2. Six to eight leaves for 10-12 months old seedlings and at least four leaves for 9 months old seedlings.
3. Collar girth of 10-12 cm.
4. Early splitting of leaves.

Note: The recovery of quality seedlings will be about 60-65 per cent. Since early germination is one of the criteria for the selection of seedlings, the storing and sowing of seed nuts should be in lots rather than in a staggered manner.

Removal of seedlings

Remove seedlings from the nursery by lifting with spade and cutting the roots. Keep the seedlings in shade and do not expose to sun. Plant seedlings as early as possible after removal from nursery. Never allow lifting the seedlings from the soil by pulling the leaves or stem.

Preparation of land and planting of seedlings

The nature of preparation of land before planting depends upon topography of land, soil type and other environmental factors. On slopes and in areas of undulating terrain, prepare the land by contour terracing or bunding. In low-lying areas and rice fields, form mounds to a height of at least 1 m above water level. In reclaimed kayal areas, planting can be done on the field bunds.

The size of pits for planting would depend upon soil types and water table. In loamy soils with low water table, pit size of 1m x 1m x 1 m is recommended. In laterite soils with underlying rock, take larger pits of size 1.2 m x 1.2 m x 1.2 m. In sandy soils, the size of pits may be 0.75 m x 0.75 m x 0.75 m. The pits may be filled up with top soil to a height 60 cm below the ground level. In low lying lands, take shallow pits and as the plant grows, raise the ground level by adding silt and sand so as to cover the entire bole of the palm. The same procedure can be adopted when planting is done on mounds or bunds. Burial of two layers of husks in the pits will be useful for moisture conservation. The husk is to be buried in layers with concave surface facing upwards.

Note: In lateritic areas, common salt @ 2 kg per pit may be applied on the pit to improve soil conditions. Common salt is to be applied about six months prior to planting.

Spacing

Spacing depends upon the planting system, soil type etc. In general, spacings are recommended under different systems in sandy and laterite soils as given in Table 22. In lateritic gravelly soils, under rainfed conditions of north Kerala, a closer spacing to accommodate 250 palms per ha is more economical.

In the hedge system of planting, the rows should be aligned in north-south direction and the seedlings planted as in the triangular system.

Time of planting

Planting the seedlings during May, with the onset of pre-monsoon rains is ideal. Under assured irrigation, planting can be done during April also. In low-lying areas, plant the seedlings in September after the cessation of heavy rains.

Shading and irrigation

For the first two years from planting, irrigate @ 45 litres of water per seedling, once in 4 days, during dry summer months. Provide adequate shade to the transplanted seedlings.

Manuring young palms

For the first three years after planting under rainfed conditions, apply fertilizers in two split doses at the rates shown in Table 23. Fertilizer requirement of adult palms is given in Table 24.

Table 22. Spacing for coconut

Planting system	Spacing	Approximate number of plants ha^{-1}
Triangular	7.6 m	198
Square	7.6 m to 9 m	170-120
Single hedge	5 m in the rows 9 m between the rows	220
Double hedge	5 m x 5 m in rows 9 m between pairs of rows	280

Table 23. Fertilizer requirement of young palms in relation to that of adult palms

Time after planting	Time of application	
	April-June	Sept-Oct.
	(Proportion of adults palm dose)	
3 months ($1/10^{th}$ of full dose)		1/10
1 year ($1/3^{rd}$ of full dose)	1/9	2/9
2 year ($2/3^{rd}$ of full dose)	2/9	4/9
3 year onwards (full dose)	3/9	6/9

Table 24. Fertilizer recommendation for coconut

	Quantity (kg/palm/annum)		
	N	P_2O_5	K_2O
1. General recommendation			
(a) Average management	0.34	0.17	0.68
(b) Good management	0.50	0.32	1.20
2. For reclaimed clayey soils (as in Kuttanad)	0.25	0.35	0.90
3. Red loam soils (southern Kerala)	0.68	0.23	0.90
4. Hybrids & high yielding palms			
(a) For irrigated areas	1.00	0.50	2.00
(b) For rainfed conditions	0.50	0.32	1.20

Note: Under irrigated conditions, the fertilizers can be applied in 3-4 equal split doses. In the case of low lying areas, apply fertilizer after water table recedes in one single dose or in two split doses as conditions permit. In all types of soils that are low in organic matter content (except reclaimed clayey soils and alluvial soils), apply organic matter @ of 15-25 kg per palm per year during June-July from the second year of planting.

Weeding and interculture

Keep the pits free of weeds by periodical weeding. Remove the soil covering the collar of seedlings. As the seedlings grow fill up the pits gradually by cutting the sides. Proper intercultivation provides control of weeds and creates soil mulch. Any tillage system (ploughing, digging, raking or forming mounds) that provides soil mulch and control weeds may be followed depending upon local conditions. For laterite, sandy and red sandy loam soils give two ploughings or diggings in May-June and September-October and one raking in January. In areas where surface run off is more, form mounds in September-October and level them in November-December.

Drought management in coconut gardens

Coconut produces nuts round the year. Therefore, adequate supply of water is essential for its proper growth. Soil moisture is essential for the absorption of nutrients by roots. Moisture stress leads to stunted growth, drooping of leaves, immature nut fall and decreased yield. Importance may be given on the following aspects so as to ward off stress:

1. Husk burial for moisture conservation

Burying of fresh or dried coconut husk around the palm is a desirable practice particularly for moisture retention. The husk can be buried either in linear trenches taken 3 m away from the trunk between rows of palms or in circular trenches taken around the palm at a distance of 2 m from the trunk. The trenches may be of 0.5 m width and depth. The husks are to be placed in layers with concave surface facing upwards and covered with soil. The beneficial effect of husk burial will last for about 5-7 years. Instead of husk, coir pith can be buried @ 25 kg per palm per year.

2. Mulching

Mulching is an effective method of conserving soil moisture. Mulch the coconut basins with green / dry leaves at the close of

northeast monsoon (October-November). Mulching also adds organic matter to the soil and reduces the soil temperature. Do not disturb soil in the coconut garden during summer months. In level lands, during rainy seasons excess water may be conserved in small trenches dug out in the plantation. In sloppy areas, land may be terraced and trenches dug across. This will facilitate maximum percolation of rainwater and water conservation. For moisture conservation, lowermost 3-5 leaves may be cut and removed. Provide adequate shade for the transplanted seedlings for 1-2 years. To minimize the sun scorch on the srunk, application of lime solution on the trunk up to a height of 2-3 m at the start of the summer season is recommended.

3. Green manure and cover crops

Green manure and cover crops recommended for cultivation in coconut gardens are:

a. Green manure crops: *Crotalaria juncea* (sunhemp), *Tephrosia purpurea* (kolinji), *Indigofera hirsuta, Pueraria phaseoloides*
b. Cover crops: *Calapagonium muconoides, Mimosa invisa, Stylosanthes gracilis*
c. Shade-cum-green manure shrub: *Tephrosia candida*

Sow cowpea seeds more towards the periphery of basins taken at a radius of 2.0m from the base of the palm for green manure during April-May with the onset of pre-monsoon rains. When a few plants start flowering, uproot the entire plants and incorporate into the soil during August-September and cover the basins with soil.

Sow green manure and cover crop seeds during April-May with the onset of pre-monsoon rains. The green manure crops should be ploughed in and incorporated into the soil during August-September. This will increase the water holding capacity of soil. *Calapagonium* can be grown either as green manure or cover crop. *Tephrosia* is especially suited for planting around seedling pits for summer shade and as a source of green manure in the rainy season.

Manuring of adult palms

The nutrient dosages recommended for adult palms are given in Table 24.

1. Under irrigated conditions, fertilizers can be applied in 3-4 equal split doses.
2. In the case of low-lying areas, apply fertilizers in one single dose after water table recedes or in two split doses as conditions permit.
3. The application of organic materials such as forest leaves, cattle manure, coir dust or coconut shredding at 10 kg per pit in the first three years and 15-25 kg thereafter will be useful to obtain better establishment of coconut palms in sandy soils and in coastal situations.
4. In situations where the available P_2O_5 status of the soil is more than 10 ppm, application of phosphatic fertilizers can be skipped for a few years until the status of P_2O_5 reaches 10 ppm.
5. For sandy and sandy loams of Onattukara and similar situations and also for hybrid palms grown in root (wilt) affected areas, apply 500 g N + 300 g P_2O_5 + 1000 g K_2O along with 500 g $MgSO_4$ per palm per year.
6. Application of $MgSO_4$ to coconut palms earlier confined to root wilt affected areas is recommended for the whole state.
7. The N:P_2O_5:K_2O recommendation given for high yielding palms is, in general, sufficient for palms yielding upto 100 nuts per year. For palms yielding more than 100 nuts per year, an additional dose of 10 g N,

5 g P_2O_5 and 15 g K_2O may be supplied for every nut exceeding 100 nuts .

8. In laterite soils, 50 per cent of the K_2O requirement of coconut can be substituted by Na_2O supplied in the form of sodium chloride.

Time, frequency and method of fertilizer application

Under rainfed conditions, apply fertilizers in two split doses, 1/3 at the time of early southwest monsoon showers in April-June and 2/3 in September-October.

Under irrigated conditions, apply fertilizers in three or four equal doses in April-May, August-September, December and February-March.

Apply lime or dolomite during April-May, magnesium sulphate during August-September and organic matter during June-July. For an adult palm 1.0 kg dolomite or 1.0 kg lime + 0.5 kg $MgSO_4$ is required per annum.

Apply fertilizers and manures in circular basins at a radius of 2.0 m from the base of the palm and 10 cm deep, opened after the onset of southwest monsoon. Split doses can be applied with irrigation water in summer months.

Recycling of palm waste

Recycling of palm waste is very much beneficial especially for maintaining the availability status of micronutrients and trace elements. Palm wastes like coconut leaves, crown waste, dried spathes, husk etc. may be deposited in a small trench of convenient length, 0.5 m to 0.75 m wide and 0.3 to 0.5 m deep at a distance of 2-2.5 m away from the base of the trunk. Fill up this trench with the palm wastes along one side of the palm (north) in one year, opposite side (south) in the next year, east in the third year and so on. This practice of organic recycling of waste has been found to improve the growth and productivity of the palms.

Intercropping and mixed cropping

Schedules for inter/mixed cropping may be drawn up based on the canopy size, age and spacing of palms. In general, palms in the age group of 8-25 years are not suitable for inter and mixed cropping. However, cereals and tapioca are recommended as intercrops in young coconut plantation upto 3-4 years. Since ginger and turmeric are shade tolerant crops with shallow roots, they can be intercropped in coconut garden even in the age group of 15-25 years. It ensures better land utilization, solar energy harvesting, efficient water use, utilization of soil nutrient resources, more returns and an insurance against crop failure. Under conditions of wider spacing i.e. beyond 7.6 m, intercropping is possible irrespective of the age of the palms.

The following crops are recommended as intercrops.

Cereals: Rice, maize

Legumes and pulses: Groundnut, horse gram, cowpea.

Tubers: Tapioca, sweet potato, yams, colocasia.

Spices and condiments: Ginger, turmeric, chilly, pepper, nutmeg, cinnamon, clove.

Fruit plants: Banana, pineapple, papaya. (Banana variety Palayankodan is recommended in the reclaimed soils of Kuttanad. Three suckers per clump have to be retained).

Beverage crop: Cocoa

Fodder grasses: Hybrid Napier, guinea grass.

In all cases, separate application of adequate fertilizers and manures to the individual crop is essential.

Casuarina *(Casuarina equisetifolia)* may be planted as single row intercrop, between rows of coconut during the first year of planting coconut seedlings in pits of size 30 cm x 30 cm x 30 cm spacing of 2m. The casuarina poles will be ready for felling/ harvesting within 5-6 years of planting.

Crop cafeteria for multiple cropping in coconut garden

Perennials: Cocoa, nutmeg, pepper, clove, lemongrass and cinnamon.

Annuals:

a. Kharif: Rice, maize, groundnut, ginger, turmeric, chilli, yams, colocasia, red gram, vegetables, sweet potato, tapioca, banana, pineapple, papaya and fodder grass.
b. Rabi: Sesame, horse gram, red gram, vegetables, cowpea, sweet potato and banana.
c. Summer: Vegetables

Irrigation

Irrigate the palms during summer months in basins around palms as shown in Table 25 & 26.

Note: In coastal sandy soils, seawater can be used for irrigation. In irrigated gardens, interruption of irrigation would lead to serious set back in yield and general condition of palms. Hence, when once started, irrigation should be continued regularly and systematically. In sandy loam soil, irrigating the crop with 500 litres of water through basin taken at 1.5 m radius at CPE value of 50 mm (approximate interval of 15 days) is most economical. Do not irrigate seedlings and very young palms with seawater.

Drip irrigation

In the traditional system of irrigation followed in coconut gardens such as flood irrigation, basin irrigation etc. irrigation efficiency is only 30 to 50 per cent due to considerable wastage of water. In addition, cost on inputs like labour and energy in adopting these systems are high. Scarcity of water and increasing cost of labour and energy are deterrents in adopting these traditional irrigation systems. Under these circumstances, drip

Table 25. Irrigation requirement of coconut

Parameters	Soil texture			
	Sandy	Sandy loam	Loam	Silty clay
Available soil moisture (cm/m)	8	12	17	21
Quantity of water / irrigation / palm in litres in a basin of 1.8 m radius	600	900	1300	1600
Frequency of irrigation (days)				
All areas in Kerala except north eastern portion of Thrissur and Palakkad districts	3-4	5	7-8	9
North eastern portion of Thrissur and Palakkad districts	2-3	3-4	5-6	6-7

Table 26. Irrigation recommendation for Coconut on Agro Ecological Unit (AEU) basis

Parameters	Agro Ecological Zones*				
	I Coastal Plains	II Midland laterites	III Foot hills	IV High hills	V Palakkad plains
Quantity of Water /irrigation / palm in litres in a basin of 1.8 m radius **	350 to 800	500 to 800	600 to 800	500 to 850	600 to 800
Irrigation interval in days**	3 to 7	3 to 7	4 to 7	3 to 9	3 to 6

**NBSSLUP classification (Nair et al., 2012).*

*** The lower values are for sandy soils and higher values are for clayey soils. Intermediate values can be chosen based on the texture of the soil with values moving towards lower side for increasing sand content and vice versa for increasing clay content. Add 30 to 40% to the above values depending upon the conveyance and application efficiencies in irrigation.*

irrigation is the most suitable system of irrigation to coconut. Some of the major advantages of drip irrigation are: it saves water, enhances plant growth and yield, saves energy and labour, most suited for soils having low water holding capacity and undulating terrain, reduces weed growth and improves efficiency of fertilizers. For coconut, generally, three to four drippers are given per palm. The water requirement for an adult palm is 40 to 50 litres per day.

D x T hybrid production

The following guidelines are suggested for large scale production of D x T hybrid seedlings. Assisted pollination should be done to get maximum hybrid nut production. As far as possible use prepotent palms as parents in the hybridization programmes.

Selection of mother palms

Palms with the following phenotypic character combination may be selected for hybridization work.

1. Nuts without ridges and having yellow, orange or red colour.
2. Palms with overlapping female and male phases.
3. Small crown and canopy compared to that of tall palm.
4. Narrow stem without any bulging at the base with close leaf scars.

Hybridization

1. Use mixed pollen from identified tall palms.
2. Emasculate the inflorescence by cutting the male flowers with scissors and stripping if necessary within 5-7 days of opening the spathe.
3. Cloth bags made of very close mesh should be used for covering the inflorescence.
4. Hairy caterpillar larvae cause serious damage by boring into the female flowers and developing buttons through stigmatic ends. The damage is more serious under bagged conditions.
5. Dusting of pollen-talc mixture in 1:9 proportion using pollen dispensers is recommended.
6. Assisted pollination for at least 3-5 days on each inflorescence till last female flower becomes receptive and fully pollinated.
7. Remove bags after the seventh day of pollination of the last female flower.

Nursery

The nuts should be harvested before it is tree-ripe and sown immediately in the bed

without storage. Nursery beds should be mulched or shaded and watered regularly and adequately.

Button shedding

The shedding of buttons in the coconut is attributed to the following reasons.

1. Pathological conditions
2. Attack of insect pests
3. Nutritional deficiencies
4. Soil and climatic variations
5. Defects in pollination and fertilization
6. Structural defects in the flower
7. Abortion of embryos
8. Limited capacity of the tree to bear fruits
9. Unfavourable conditions such as deficit of moisture, waterlogging and lack of aeration.

The causes of button shedding may be identified and appropriate remedial measures adopted.

Plant protection

Pests

Rhinoceros beetle *(Oryctes rhinoceros)*

Symptoms

The adult beetle bores into the unopened fronds and spathes. The attacked frond when fully opened shows characteristic triangular cuts.

Management

1. Provide field sanitation to prevent breeding of beetles.
2. Hook out the beetles from the attacked palms by using beetle hook.
3. The topmost three leaf axils around the spindle may be filled with any of the following mixtures as a prophylactic measure:
   a. Application of 250 g neem cake or marotti (*Hydnocarpus wightina*) cake mixed with equal volume of sand in the innermost 2-3 leaf axils. This treatment is to be done twice, ie., during April-May before the onset of south-west monsoon and during September-October after the south-west monsoon.
   b. Naphthalene balls 12.0 g (approx. 4 nos.) in the innermost 2 leaf axils and covered with fine sand, once in 45 days.
   c. Incorporation of *Clerodendron infortunatum* @ 10 per cent *w/w* basis in the cowdung pit and/or manure pits.
4. Crown cleaning followed by the application of 20g Cartap hydrochloride 4G or Fipronil 0.3G or carbosulfan 6G mixed with 200 g sand in the innermost 2-3 leaf axils. This treatment has to be done thrice during January, May and September.
5. Release *Baculovirus oryctes* infected adults @ 10-15 ha^{-1} to bring down the pest population.
6. Inoculation of breeding sites with entomopathogenic fungus *Metarrhizium anisopliae* (@ 5 x 10^{11} spores / m^3) var major causes mortality to the grubs.

Red palm weevil

(Rhyncophorus ferrugineus)

Symptoms

The diagnostic symptoms are the presence of holes on the stem, oozing out of a viscous brown fluid and extrusion of chewed up fibrous matter through the hole, longitudinal splitting of leaf base and wilting of central shoot. Sometimes the gnawing sound produced by the feeding grubs inside will also be audible.

Management

1. Field sanitation should be given prime importance.
2. Avoid making steps or any other injury on the tree trunks to reduce the loci of infestation.

3. Leaf axil filling as suggested in the case of rhinoceros beetle will be useful against the red palm weevil also.
4. When green leaves are cut from the palms, stumps of not less than 120 cm may be left on the trees in order to prevent successful inward movement of the grubs through the cut end.
5. In attacked palms, observe for the bore-holes and seal them except the top most one. Through the top most hole, pour 1 per cent carbaryl or 0.15 per cent trichlorphon suspension @ one litre per palm, using a funnel.
6. When the pest infestation is through the crown, clean the crown and slowly pour the insecticidal suspension.
7. As an alternative, apply 1 per cent DDVP as a curative measure.
8. Coconut log traps with fermenting toddy or pineapple or sugarcane activated with yeast or molasses can be set in coconut plantation to attract and trap the free floating population of red palm weevil. Incorporate any of the insecticide to each trap to kill the weevils trapped.
9. Use of pheromone trap for attracting and killing adult weevils @ one trap per 2 ha.

Leaf eating caterpillar *(Opisina arenosella)*

Symptoms

The caterpillar feeds on green matter from the lower leaf surface, remaining within galleries of silk and frass. The attack will be severe during summer months from January-May.

Management

1. As a prophylactic measure, the first affected leaves may be cut and burnt during the beginning of the summer season.
2. Arrange for the release of larval / pupal parasitoids, *Goniozus nephantidis, Elasmus nephantidis* (brown species) and *Brachymeria nosatoi.*
3. Trunk release of *G.nephantidis* @ 10 nos/ plam (4 to 6 releases) is effective for the management of leaf eating caterpillar.
4. When infestation is very severe and if the biocontrol is not likely to be effective, spray the undersurface of the fronds with dichlorvos 0.05 per cent, malathion 0.1 per cent, quinalphos 0.05 per cent or phosalone 0.07 per cent.
5. When the infestation is very severe and if the biocontrol is not likely to be effective, spray the under surface of the fronds with dichlorvos 0.05 per cent, malathion 0.1 per cent, quinalphos 0.05 per cent, phosalone 0.07 per cent, Flubendiamide 39.35 SC (0.2 ml l^{-1}), Chlorantraniliprole 18.5% SC (0.5 ml l^{-1}). Spinosad 45 SC (0.4 ml l^{-1}) is also effective.

Note: Application of the insecticides should be followed by liberation of larval and pupal parasites from the 21st day.

Cockchafer beetle

(Leucopholis coneophora)

Symptoms

The soil inhabiting white grubs cause damage to the roots of coconut palm. The attack is common in sandy tracts. The infested palms turn pale yellow and there will be considerable reduction in yield.

Management

1. Collection and destruction of adults during the monsoon period from adjacent vegetation (in the evening).
2. Plough or dig the infested soil synchronizing with pre-monsoon showers.
3. Drench the soil with chlorpyrifos 0.04 per cent suspension. The treatment should be given twice, first during April-May after the receipt of pre-monsoon showers and second during the month of September.

Note: Wherever possible, light traps may be set up to attract and trap adult beetles.

Coreid bug *(Paradasynus rostratus)*

Symptoms

The attacked buttons become deformed with characteristic crevices on the husk below the perianth with gum exudations and the tender nuts become barren.

Management

Apply 0.1 per cent Dimethoate 0.05% on the newly opened inflorescence after the receptive phase of the female flowers and spray the entire crown excluding the leaves and older bunches.

Note: The insecticide may be applied according to the severity of infection in a need-based manner.

Coconut eriophyid mite

(Aceria guerreronis)

It is a microscopic worm like mite infesting young buttons colonizing under the perianth.

Symptoms

The earliest symptoms on 2-3 months old button is pale yellow triangular patches seen below the perianth. Later, these patches become brown. Severely affected buttons may fall. As the buttons grow, brown patches lead to black necrotic lesions with longitudinal fissures on the husk. Uneven growth results in distortion and stunting of nuts leading to reduction in copra yield. In severe cases, the losses are compounded because the quality of fibre is reduced and distorted nuts increase the labour requirements for dehusking.

Management

1. Collect and destroy all the fallen buttons of the affected palm.
2. Apply 2 per cent neem oil + garlic emulsion or commercial neem formulation azadirachtin 0.004 per cent (Neemazal T/S 1 per cent @ 4 ml per litre of water) or micronized wettable sulphur 0.4 per cent in the crown on young bunches. When rocker sprayer is used 1.0 to 1.5 litres of spray fluid per palm is required. If a hand sprayer is used, the spray solution required may be about 500 to 750 ml. Spraying has to be done on second to seventh bunches from top avoiding unpollinated inflorescence. Care should be taken to see that spray fluid reaches the perianth region of third, fourth and fifth bunches since these bunches harbour maximum number of mites. Three rounds of spraying are recommended in a year viz., March-April before the onset of southwest monsoon, in August-September during the dry spell between the southwest and northeast monsoons and in December-January after the northeast monsoon so that all the emerging bunches in the vulnerable stage receive one round of spraying. Rational rotation of the above pesticides may be adopted to avoid chances of resistance.

Preparation of neem oil + garlic emulsion (2 per cent)

To prepare 10 litres of 2 per cent neem oil + garlic emulsion, 200 ml neem oil, 200 g garlic and 50 g ordinary bar soap are required. Slice the bar soap and dissolve in 500 ml lukewarm water. Grind 200 g of garlic and take the extract in 300 ml of water. Pour the 500 ml soap solution in 200 ml neem oil slowly and stir vigorously to get a good emulsion. Mix the garlic extract in the neem oil + soap emulsion. Dilute this 1 litre stock solution by adding 9 litres of water to get 10 litres of 2 per cent neem oil + garlic emulsion.

As per the recommendation of the National Level Steering Committee, a holistic approach has to be adopted in the management of the coconut eriophyid mite. Hence, in addition to the plant protection measures mentioned above, the following measures can be adopted:

1. Improving nutrient status by applying organic manure at the rate of 50 kg and neem cake 5 kg per palm per year. Also apply the recommended dose of fertilizers in two split applications.
2. Growing compatible intercrops / mixed crops.
3. Providing adequate irrigation.

Mealy bug

Symptoms

Mealy bugs infest the unopened heartleaf and inflorescence. As a result, the leaves become highly stunted, suppressed, deformed and present a crinkled appearance. It is often confused with the leaf rot symptoms. The affected inflorescences are malformed and do not open. Even if they open, they do not bear nuts.

Button mealy bugs colonize under the perianth lobes of tender nuts. Infested nuts harbouring gravid mealy bugs remain on the spadix, which serve as inoculum for further spread.

Management

Remove and destroy all dried up inflorescence and unproductive buttons. Apply non-residual phosphatic insecticide dimethoate 0.05 per cent or quinalphos 0.05 per cent at the site of infestation. Neem garlic emulsion 2 per cent applied on infested bunches checks button mealy bugs.

Rodents

Rats damage tender nuts by forming characteristic holes. Shed nuts can be seen at the base of the palm.

Place wax blocks of 0.005 per cent bromadiolone in coconut crown of the infested palms at 3 to 4 days interval till the bait is no more consumed.

Diseases

Phytophthora diseases

Phytophthora palmivora has been found to affect seedlings and adult palms causing bud rot and immature nut fall commonly known as mahali.

Bud rot

Symptoms

Palms of all age are liable to be attacked but normally young palms are more susceptible, particularly during monsoon when the temperature is low and humidity is very high. In seedlings, the spear leaf turns pale and comes off with a gentle pull. In adult palms, the first visible symptom is the colour change of the spear, which becomes pale and breaks at the base and hangs down. The tender leaf base and soft tissues of the crown rot into a slimy mass of decayed material emitting a foul smell. The rotting slowly progresses downwards, finally affecting the meristem and killing the palms. This is accompanied by drooping of successive leaves. Even then, nuts that are retained on the palm may grow to maturity. The disease proves fatal if not checked at the early stages, before damage of the bud.

Management

1. In early stages of the disease (when the heartleaf starts withering) cut and remove all affected tissues of the crown. Apply Bordeaux paste and protect it from rain till normal shoot emerges.
2. Burn all disease-affected tissues removed from the palm.
3. Spray 1.0 per cent Bordeaux mixture on spindle leaves and crown of disease affected as well as neighbouring palms, as a prophylactic measure. Palms that are sensitive to copper containing fungicides can be protected by mancozeb. Small, perforated sachets containing 2 g of mancozeb may be tied to the top of leaf axil. When it rains, a small quantity of the fungicide is released from the sachets to the leaf base, thus protecting the palm.
4. Adopt control measures for rhinoceros beetle.
5. Provide adequate drainage in gardens.
6. Adopt proper spacing and avoid over crowding in bud rot prone gardens.

Mahali

Symptoms

Shedding of female flowers and immature nuts are the common symptoms of the disease. Lesions appear on the young fruits or buttons

near the stalk, which later lead to the decay of the underlying tissues and endosperm.

Management

Spray 1.0 per cent Bordeaux mixture or copper oxychloride preparation (2 g l^{-1}) on the crown of palms, once before the monsoon and once or twice later on at intervals of 40 days.

Root (wilt) disease

Symptoms

The characteristic symptom is the flaccidity of leaflets. Yellowing of older leaves, necrosis of leaflets and deterioration and decay of root system are other salient features of the disease. The leaflets curve inwardly to produce ribbing so that the whole frond develops a cup like appearance. Abnormal shedding of buttons and immature nuts are also noticed.

Management

Coconut root (wilt) is a non-lethal debilitating disease and the affected palms survive for a long period giving a reasonably good yield. The root (wilt) affected palms are susceptible to diseases like leaf rot and pests like rhinoceros beetle and red palm weevil. So there is a chance of confusing the pests and disease symptom with the root (wilt) disease. Negligence on the management aspects aggravates the malady. Efficient management of palms suspected to be affected by coconut root (wilt) disease demands control of all pests and diseases and imparting natural resistance and health to the palms through proper manuring and agronomic practices. A package of management practices for the effective management of root (wilt) disease is given below:

1. Rogue out palms that are affected severely by root (wilt) and yield less than 10 nuts / palm / year and those, which have contracted the disease before flowering. Replant with disease tolerant material / high yielding hybrids (Chandrasankara).
2. Apply fertilizers for coconut palms in average management at the rate of 0.34 kg N, 0.17 kg P_2O_5 and 0.68 kg K_2O / palm / year in the form of urea, rock phosphate and muriate of potash, respectively. For palms under good management, fertilizers may be given @ of 0.5 kg N, 0.32 kg P_2O_5 and 1.2 kg K_2O / palm / year.
3. In addition to the above, apply 50 kg cattle manure or green manure and 1 kg of lime/palm/year. Magnesium may be supplied @ 500 g MgO per palm per year in the Onattukara region (sandy soil) and 100 g MgO in the remaining areas. The cheapest source of MgO is magnesite ($MgCO_3$). The magnesium in magnesite is acid soluble. Hence it may be preferred in acid soils.
4. Growing green manure crops like sun hemp, sesbania, cowpea and calapagonium in the coconut basin and their incorporation *in situ* is beneficial as the practice reduces the intensity of the root (wilt) and increases the nut yield. The ideal green manure crops for the sandy and alluvial soils are cowpea and sesbania, respectively.
5. Under rainfed conditions, apply fertilizers in two splits, 1/3rd at the time of early southwest monsoon and 2/3rd before the northeast monsoon. Under irrigated conditions apply fertilizers in three equal splits (April-May, August-September and December-January).
6. Apply fertilizers and manures in 10 cm deep circular basins at a radius of 2 m from the bole of the palm.
7. When the crop is grown under the bund and channel system, desilt the channel and strengthen the bunds during summer months.
8. Follow strictly all the prescribed prophylactic measures against leaf rot

disease, red palm weevil, rhinoceros beetle etc. so as to ensure that the palms are not debilitated. To maintain the productivity of the palms, prophylactic measures are of great importance.

Leaf rot

Symptoms

The first symptom is the appearance of water-soaked brown lesions in the spear leaves of root-wilt affected palms. Gradually these spots enlarge and coalesce resulting in extensive rotting. As the leaf unfurls, the rotten portions of the lamina dry and get blown off in wind, giving a 'fan' shape to the leaves. Some times, the symptom becomes very acute and the spear fails to unfurl.

This disease is a fungal complex initiated predominantly by *Colletotrichum gloeosporioides, Exserohilum rostratum* and *Fusarium* spp.

Management

1. Remove the rotten portions from the spear and the two adjacent leaves.
2. Pour 300 ml of fungicidal solution at the base of the spear. This can be prepared by mixing hexaconazole 5 EC 2 ml or mancozeb 3 g in 300 ml water.
3. Treat the top two leaf axils with insecticide preparation. This can be prepared by mixing cartap hydrochloride 20 g with 200 g sand.
4. Spray crowns and leaves with 1per cent Bordeaux mixture or copper oxy chloride formulations (2 g l^{-1}) or mancozeb (3 g l^{-1}) in January, April-May and September. While spraying, care has to be taken to spray the spindle leaf.
5. After crown cleaning (removal and destruction of affected parts) in coconut palms pour 300 ml of any of the following bio-control agent at the base of spear leaves twice a year (April-May and September-October) 1. *Pseudomonas fluorescens* (20 g l^{-1}) or 2. PGPR mix II (20 g l^{-1}).

Stem bleeding

Symptoms

Exudation of the reddish brown liquid through the growth cracks mostly at the basal part of the trunk and bleeding patches higher up in the trunk are characteristic symptoms. One or more lesions, lying close by, may coalesce to form large patches. The liquid that oozes out dries up and turns black. The tissues beneath the bleeding points decay and become yellowish. The lesions spread upwards as the disease progresses. In advanced stages, the leaf size reduces leading to reduction in crown size. The rate of leaf production slows down. The production of bunches is affected and nut shedding takes place. The trunk gradually tapers towards the apex. The progress of the disease is faster during July to November.

Causal organism

The fungus, *Thielaviopsis paradoxa* is the causal agent. Growth cracks on the trunk, severe summer followed by heavy down pour, water stagnation, imbalance in nutrition, excess salinity and stress can act as predisposing / aggravating factors.

Management

1. Chisel out completely the affected tissues and paint the wound with Bordeaux paste. Apply coal tar after 1-2 days.
2. Destroy the chiseled materials by burning. Avoid any mechanical injury to trunk.
3. Apply neem cake @ 5 kg per palm in the basin along with other organics.
4. Apply hexaconazole @ 25 ml in 25 litre of water as soil drenching once in four months.
5. Swabbing the lesions on the trunk with a paste of *Trichoderma viride* talc based

formulation (100 g 100 ml^{-1}) along with application of lime or dolomite @ 1kg $palm^{-1}$ during April-May and application of neem cake (5kg $palm^{-1}$) + organic manure enriched with *Trichoderma viride* (1%) during September-October.

Grey blight

This is caused by the fungus *Pestalotia palmarum.*

Symptoms

Symptoms appear in the mature leaves of the outer whorl as yellow specks encircled by a greyish band which later become greyish white. The spots coalesce into irregular necrotic patches causing extensive leaf blight. In advanced stages, the tips and margins of the leaflets dry and shrivel giving a burnt appearance.

Management

Remove severely affected older leaves and burn. Spray the trees with 1 per cent Bordeaux mixture or propiconazole @ 1 ml l^{-1}.

Tanjore wilt

Symptoms

This disease is of recent occurrence in many parts of Kerala, especially in the districts of Palakkad, Malappuram, Thrissur, Kollam, Thiruvananthapuram and Wayanad. Middle aged palms were seen fatally affected. The characteristic symptom of the disease is the rotting of the basal portion of the stem. The bark turns brittle and often gets peeled off in flakes, leaving open cracks and crevices. The internal tissues are discoloured and disintegrated, emitting a bad smell. Mild bleeding occurs on the basal region. The tissues on the bleeding spots are soft to touch. Extensive damage of the root system following root rotting has been observed. Ultimately the palm dies off.

Management

1. Apply organic manure @ 50 kg $palm^{-1}$.
2. Apply neem cake @ 5 kg $palm^{-1}$ $year^{-1}$.
3. Reduce fertilizer application to one-fourth of the recommended dose.
4. Drench the basin with 40 litres of any copper fungicide to soak soil up to 15 cm depth at quarterly intervals.
5. Avoid flood irrigation in order to prevent the possible spread of the pathogen through soil.
6. Isolate the affected palm from the healthy ones by digging a trench of size 1 m deep and 50 cm wide, 1.5 m away from the bole of the infected palm.
7. Avoid growing leguminous crops in and around the garden.

Safe storage of copra

Copra obtained from commonly cultivated varieties / cultivars is attacked by various insect pests in store. Among these ham beetle, *Necrobia rufipes* and saw toothed grain beetle, *Oryzaphilus surinamensis* are of major importance, which can cause more than 15 per cent loss to copra when stored for more than six months.

Following precautions are to be taken for the safe storage of copra for more than three months:

1. Dry the produce to 4 per cent moisture content.
2. Avoid heap storage, which causes maximum damage.
3. Store copra in netted polythene bags or gunny bags.

Shell fired copra dryer

A new type of dryer working on indirect heating and natural convection principles using coconut shell as fuel has been designed and developed by CPCRI. The overall dimension

of the dryer is 2.25 m length, 1.5 m breadth and 1.5 m height. The capacity of the dryer is 1000-1200 nuts per batch. The dryer has two heating chambers which are arranged in parallel. Specially developed rolling in type of fuel trays are used for burning the fuel. The dryer consists of a drying chamber, a burning chamber, a plenum chamber and ventilation holes. Each full tray produces heat for 6 hours, with a temperature of about 80-82°C. Generally after 6 hours, when the temperature drops below 60°C, the fuel trays are removed from the dryer, cleaned and reloaded with fuel, refired and replaced into respective burning chambers. About 4 loads of fuel are required with a capacity of 80 shells per tray to dry the copra to about 6.25 per cent moisture content. The total drying time is 24 hours. The cost of the dryer is approx Rs.35,000/-. Farmers and entrepreneurs interested to purchase this dryer can contact, Agricultural Technology Information Centre (ATIC), CPCRI, Kasaragod, for further details.

GROUNDNUT (*Arachis hypogaea*)

Groundnut can be cultivated as a floor crop in coconut gardens, as an intercrop with tapioca and as a catch crop after second crop paddy with irrigation. The crop is grown best in sandy loam and loam soils.

Season

Rainfed : May-June to September-October
Irrigated : January to May

Varieties

	Duration (days)
TMV-2 (bunch type)	110
TMV-7 (bunch type)	110
TG-3 (bunch type)	100-110
TG-14 (bunch type)	105-115
Spanish Improved	100-110
Sneha (bunch type)	Early
Snigtha (bunch type)	Early

Note: TG-3, TMV-2 and TMV-7 are recommended as intercrops in coconut gardens.

Seeds and sowing

Pure crop	100kg kernels ha^{-1}
Intercrop in coconut	80 kg kernels ha^{-1}
Intercrop in tapioca	40-50 kg kernels ha^{-1}

Plough the field three or four times into a fine tilth. Sow the seeds by dibbling in ploughed furrows at a spacing of 15 cm x 15 cm. For seed treatment the rhizobial culture is recommended.

Manuring

Cattle manure or compost	2 t ha^{-1}
Lime	1-1.5 t ha^{-1}
Fertilizers $N:P_2O_5:K_2O$	10:75:75 kg ha^{-1}

Apply entire quantity of cattle manure or compost and recommended quantity of fertilizers as basal dressing and incorporate well into the soil. Apply lime at the time of flowering of the crop and mix with the soil by light hoeing or raking.

Irrigation and interculture

Irrigate the crop once in 7 days. Weed the crop 10-15 days after germination of seed by light hoeing. Give another light hoeing or

Pests

The hard scale *Saissetia nigra* occurs on the pencil thick branches and desaps the tissues. The infested shoots invariably develop sooty mould cover. It can be controlled by spot spraying with quinalphos 0.025 per cent.

Diseases

Leaf spot and die back *(Colletotrichum gloeosporioides)*

Sunken spots surrounded by a yellow halo are the initial symptoms. Subsequently the central portion of the necrotic region drops off resulting in shot hole symptoms. Dieback symptoms are also observed in some of the mature branches. On young seedlings drying of the leaves and subsequent defoliation are seen. The disease can be controlled by spraying 1 per cent Bordeaux mixture two or three times during rainy season.

Fruit rot

This is caused by *Colletotrichum gloeosporioides* and *Botryodiplodia theobromae*. Water soaked lesions are seen on the fruits, the tissues of which become discoloured and disintegrated. Premature splitting of the pericarp and rotting of mace and seed are the main symptoms of the disease. The internal tissues are found rotten. The fallen fruits become enveloped with the growth of the organism. The disease can be controlled by spraying 1 per cent Bordeaux mixture.

Other diseases: Phytophthora leaf fall, white thread blight, horse hair blight, sooty mould and algal leaf spot.

BLACK PEPPER (*Piper nigrum*)

Pepper requires a warm and humid climate. Though an annual rainfall of 250 cm is ideal for the proper growth of the crop, it can also come up well in low rainfall areas, if the pattern and distribution of rainfall are conducive. About 70 mm of rainfall within a period of 20 days may be sufficient for triggering of flushing and flowering process in the plant, but once the process is set on, there should be continuous, rainfall until fruit development starts. Any dry spell, even for a few days, within this critical period will result in substantial reduction of yield. Very long spells of dry weather are unfavourable for the crop growth.

The plant tolerates a minimum temperature of 10°C and maximum of 40°C, the optimum being 20 - 30°C. It can be grown from sea level upto an altitude of 1200 m.

Pepper prefers a light porous and well-drained soil rich in organic matter. Water stagnation in the soil, even for a very short period, is injurious for the plant. So, heavy textured soils in locations where drainage facilities are inadequate should be avoided.

Varieties

Improved varieties : Panniyur-1, Panniyur-2, Panniyur-3, Panniyur-4, Panniyur-5, Panniyur-6, Panniyur-7, Panniyur-8, Vijay, Subhakara, Sreekara, Panchami, Pournami, Girimunda and Malabar Excel. IISR Sakthi and IISR Thevam are tolerant to Phytophthora foot rot.

Local varieties: Karimunda, Neelamundi, Kottanadan, Kuthiravally, Arakulam Munda, Balankotta and Kalluvally.

Selection of site

Sites with slight to moderate slope are ideal for pepper cultivation, as they promote drainage. Slopes facing south are to be avoided as far as possible. When such slopes are to be used for cultivation, the young plants may be sufficiently protected from the scorching sun during summer.

Selection of mother plants

Cultivate varieties, which are proven to be highly productive. Select mother plants, which give regularly high yields and possess other desirable attributes such as vigorous growth, maximum number of spikes per unit area, long spikes, close setting of berries, disease tolerance etc. Selected mother plants should be in the age group of 5-12 years. Mark and label selected mother plants in October-November.

Raising of rooted cuttings

Pepper is propagated vegetatively from cuttings. Select runner shoots produced at the base of mother plants and keep them coiled and raised to prevent from striking roots in the soil. Separate them from the vines in February-March.Very tender and too hard portions of the shoots are to be avoided for planting. The shoots are cut into pieces with 2-3 nodes in each. Two node semi-hard wood cuttings are to be planted for rooting of pepper cuttings. Leaves, if any, are to be clipped off leaving a small portion of the petioles on the stem. Satisfactory rooting and survival of cuttings (over 70 per cent) could be achieved even without any hormone treatment. Plant the cuttings in polythene bags filled with potting mixture. The potting mixture is prepared by mixing two parts of fertile topsoil, one part of river sand and one part of well rotten cattle manure. Substituting granite powder (a waste material from stone quarries) for sand in conventional potting mixture (2:1:1) is good for growth of pepper cuttings and is economical. Recommended for black pepper nurseries for large scale multiplication. Solarized potting mixture supplemented with nutrient solution (urea, superphosphate, MOP and magnesium sulphate in 4:3:2:1 ratio) and fortified with biocontrol consortia promotes growth and helps in production of disease free rooted cuttings. When polythene bags are used, sufficient number of holes (16-20) may be provided at the base to ensure good drainage. The cuttings should be planted at least one node deep in the soil. The cutting after planting should be kept under good shade. In large nurseries, pandals are to be constructed for this purpose. The cuttings are to be well protected from direct sunlight and frequent watering is recommended in the nursery to maintain a humid and cool atmosphere around the cuttings. Watering 2-3 times a day is sufficient. Heavy watering, which makes the soil slushy and causes water logging is to be avoided.

Partially composted coir pith and vermicompost (75:25) enriched with *Trichoderma* (in talc formulation, 10^7 cfu g^{-1} at the rate of 10 g kg^{-1}) is found to be an ideal potting medium for black pepper nursery for healthy planting material production using plug-trays (cell dimension of 7.5 x 7.5 x 10.0 cm) compared to conventional multiplication. It reduces the incidence of soil born diseases and aids easy transportation of seedlings.

Serpentine method of propagation

Three node cuttings planted in polythene bags are kept in a corner of the nursery. When the plant develops two leaves they are trailed horizontally in polythene bags containing potting mixture kept below each tender node.

Each node will be pressed into the mixture with polythene bags with 'V' shaped midribs of coconut leaves. As new shoots arise these will be trailed horizontally in polythene bags containing potting mixture. Upward growth of cutting is not arrested. Once twenty nodes get rooted first 10 bags in the rooted nodes will be separated by cutting at the inter nodes. The inter nodal stub will be pushed back into the potting mixture. These stubs also produce a second root system. Daily irrigation is to be given using a rose can. After three months it will be ready for planting in the main field. On an average 60 cuttings will be obtained in a year by this method from each mother cutting. Recommended in black pepper nurseries for large scale multiplication.

Field planting

Planting of standards is to be taken up in April-May with the onset of pre-monsoon showers. Karayam or Killingil (*Garuga pinnata*), *Matti/Pongalyam (Ailanthus* sp.), subabul (*Leucaenea leucocephala*) etc. are suitable standards for growing pepper. *Acacia auriculiformis* and *Artocarpus heterophyllus* are two multipurpose support tree species suitable for block cultivation of black pepper (*Piper nigrum* L.). Pepper yields are promising when trees are grown in high density (3 x 3m spacing; 1111 trees ha-^{1}) block plantations. Both the tree species provide additional economic returns through timber yield at final harvest. Because of prevalence of Erythrina gall wasp avoid using Erythrina as standard. Only species tolerant to Erythrina gall wasp is *Erythrina variegata* (heavily throny). In high altitude areas, dadap (*E. lithosperma*) and Silver oak (*Grevillea robusta*) can be successfully used as standard for pepper. Seedlings of subabul and silver oak are to be planted 2-3 years before planting pepper. The cuttings of standards are to be planted in narrow holes of 40 to 50 cm depth. The spacing recommended is 3 m x 3 m on plain lands and 2 m between plants in rows across the slope and 4 m between rows on sloppy lands. The soil should be pressed well around the standards to avoid air pockets and keep the standards firm in the soil.

For planting pepper, prepare pits on the northern side of standards, 15 cm away from it. The pit size should be 50 cm x 50 cm x 50 cm. Fill the pits with a mixture of top soil and compost or well rotten cattle manure @ 5 kg per pit and 50 g of *Trichoderma*. With the onset of south west monsoon in June-July, plant 2 rooted cuttings in the pits at a distance of about 30 cm away from the standards. Press the soil around the cuttings and form a small mound slopping outward and away from the cuttings to prevent water stagnation around the plants. The growing portions of the cuttings are to be trailed and tied to the standards. Provide shade to the plants if the land is exposed and if there is a break in the rainfall. When pepper is trailed on arecanut, plant the cuttings 1.0 m away from palm and 1.5 m when coconut is used as the support. Trail the pepper vines on a temporary stake for 1-2 years. When they attain sufficient length to reach the tree trunk, remove the stake without causing damage to the vines and tie the pepper plants on to the tree trunk and trail them on it.

Management after planting

If the terrain of the land is sloppy or uneven, carry out contour bunding or terracing to prevent soil erosion. Carry out digging around the standards and vines at a radius of about 1 m from the base or in the entire plantation, twice during the year, the first at the onset of southwest monsoon and the second towards the end of northeast monsoon. Weeding around the plants is to be done according to necessity. However, in foot

rot affected gardens, digging should be avoided and weeds are removed by slashing. In the early stages, tie the vines to the standards, if found necessary.

Where pepper is grown in large areas, growing of cover crops like *Calapagonium muconoides* is recommended. When such cover crops are grown, they are to be cut back regularly from the base of the plants to prevent them from twining along with the pepper vines. Lowering of vines after one year's growth will promote lateral branch production.

Intercropping of pepper gardens with ginger, turmeric, colocasia and elephant foot yam is advantageous. Banana as an intercrop in yielding gardens reduces pepper yield. Therefore, this is not recommended beyond three to four years after planting of pepper vines. However, in the early years, banana provides shade to the young plants and protects them from drying up during summer months.

When pepper is grown in open places, shading and watering of the young seedlings may be done during summer months for the first 1 to 3 years according to necessity. The young plants may be completely covered with dry arecanut leaves, coconut leaves or twigs of trees until summer months are over. Mulching the basins of pepper vines during summer months is highly advantageous. Saw dust, arecanut husk and dry leaves are suitable mulching materials. Removal of unwanted terminal shoot growths and hanging shoots should be done as and when necessary.

Prune and train the standards in March-April every year to remove excessive overgrowth and to give them a proper shape. The effective height of the standard is to be limited to about 6 m. A second pruning of the standards may be done in July-August, if there is excessive shade in the garden. PGPM soil application (50 g/vine) and spraying (2%) with consortium of PGPM helps to prevent phytophthora foot rot infestation.

Underplanting

After regular bearing for about 20 years, the vines of most varieties start declining in yield. The age of decline in yield varies with variety and agroclimatic and management factors. So underplanting should be attempted at about 20 years after planting or when a declining trend in yield appears. The old and senile vines can be removed 3-5 years after underplanting depending upon the growth of the young vines.

Manuring

Manuring for pepper vines is to be done in basins taken around the plant, 10-15 cm deep and 30-40 cm radius, depending upon the growth of the plants. Apply cattle manure / compost / green leaves @10 kg/ plant / annum just at the onset of southwest monsoon and cover lightly with soil. It is desirable to apply lime at the rate of 500 g/vine in April-May, with the receipt of pre-monsoon showers, in alternate years.

Apply 10 kg FYM + 500g Neem cake + 500 g ash+ 2 kg vermicompost with biofertilizer – *Azospirillum & P solubilizing bacteria* (20g).

Apply PGPR Consortia (50 g) consisting of 2 *spp*. of *Micrococcus* and I *sp*. of *Enterobacter* per vine which helps in growth promotion and disease suppression.

Recommended nutrient dosage for pepper (3 years and above) is:

$N:P_2O_5:K_2O$ *g/vine/year*

50:50:150 (general recommendation)

50:50:200 (for Panniyur and similar areas)

140:55:275 (for Kozhikode and similar areas)

Note: Apply 1/3 dose for one year old plants and 1/2 dose for two year old plants.

The fertilizers may be applied in two split doses, the first in May-June with the receipt of a few soaking rains and the second in August-September. Apply fertilizers in a circle of radius 30 cm around the vine in the case of plants trailed on erythrina (*Nadan murukku*) or teak pole (dead standard) soil application of zinc @ 6 kg ha^{-1} as zinc sulphate or foliar spray of Zn @ 0.5 per cent during flowering and pin head stage of black pepper is recommended in zinc deficient areas of black pepper cultivation for increasing the yield and quality. Application of molybdenum @ 1 kg ha^{-1} is recommended for areas deficient in soil molybdenum availability.

Irrigation

Irrigating pepper plants of Panniyur-1 variety at IW/CPE ratio of 0.25 from November / December till the end of March and withholding irrigation thereafter till monsoon break, increases pepper yield by about 50 per cent. The depth of irrigation recommended is 10 mm (100 litres of water per vine per irrigation at an interval of about 8-10 days under Panniyur conditions). The water is to be applied in basins taken around the plants at a radius of 75 cm. The basins may be mulched with dry leaves or other suitable materials. The irrigation schedule for AEUs of Wayanad district is given below.

Parameters	Agro Ecological Units of Waynad District*		
	AEU-15 Northern High hills (Sandy Clay loam)	AEU-20 Wayanad Central plateau (Clay loam to Clay)	AEU-21Wayanad Eastern plateau (Sandy to Sandy loam)
Quantity of Water/ irrigation/ vine in litres in a basin of 0.75 m radius**	25	35	20
Irrigation interval in days***	5 to 7	7 to 10	4 to 7

* NBSSLUP classification (Nair *et al.,* 2012).

** Add 30 to 40% to the above values depending upon the conveyance and application efficiencies.

*** Shorter irrigation interval is for the months from March to May

Bush pepper

For production of bush pepper, two to four node semi hard wood lateral branches with orthotropic shoot from where they arise are to be collected and planted in the nursery for rooting during June - July. Well rooted plants are used for field planting. The rooted cuttings are to be planted in pits or pots. Fertilizers can be applied @ 1.0, 0.5 and 2.0 g/

pot of N, P_2O_5 and K_2O respectively at bimonthly interval. Alternatively, application of 15 g groundnut cake or 33 g of neem cake can also meet the N requirement of the crop. The bushy nature of the plant will have to be ensured by proper pruning of the viny growth. The potted plants are to be kept preferably under partial shade. It is necessary that re-potting is carried out after every two years.

Irrigating black pepper vines with 8 litres of water through drip per day during Oct - May enhances yield and quality in bush pepper with high B:C ratio. Recommended for bush pepper grown as intercrop in coconut gardens.

Plant protection

Pests

For the control of pollu caused by the flea beetle *Longitarsus nigripennis*, spray any one of the following insecticides namely, dimethoate or quinalphos at 0.05 per cent concentration. The spraying is to be given at the time of spike emergence (June-July), at berry formation (September-October) and once again at berry maturing stage, if needed. It can also be controlled by spraying cypermethrin 0.01 per cent twice, first at the berry formation stage and the second one-month after the first spray (Sept.-Oct.).

For controlling pepper leaf gall thrips, dimethoate 0.05 per cent may be used.

Three different types of scale insects are found infesting black pepper in high ranges of Idukki district. They are black pepper mussel scale (*Lepidosaphes piperis* Gr.) infesting all parts of vines, coconut scale (*Aspidiotus destructor* Sign) feeding from undersurface of leaf and soft scale *Marsipococcus marsupiale* Gr. confining to upper leaf surface. Infestation by mussel scale causes significant loss of yield as it affects all parts of plant including berries.

Two sprayings of dimethoate 0.05 per cent at fortnightly intervals after the harvest of berries effectively control black pepper mussel scale.

Soft scale (*Lecanium* sp.) is occasionally found to infest the foliage and vines at higher elevations. This scale insect can be controlled by spraying quinalphos 0.05 per cent.This treatment will be adequate to control the mealy bugs also. Root mealy bugs can be controlled by drenching the basins of vines with chlorpyriphos 0.075 per cent. Adequate precaution has to be taken to ensure that the insecticide solution reaches the root zone of the vines. Many of the vines infested by root mealy bugs are also likely to be infected with *Phytophthora* and nematodes. For controlling hard scale, spot application of dimethoate 0.1 per cent is recommended.

Two sprays of Azadiractin 5000 ppm at 15 days interval after the incidence of scale insect.

Thiamethoxam 0.013% two sprays at an interval of 15 days after incidence of scale insect is also recommended.

Top shoot borer can be controlled by spraying dimethoate (0.05 per cent) on the tender shoots and flushes. The spraying has to be repeated to protect newly emerging shoots and flushes.

For control of the burrowing nematode *Radopholus similis* and the root knot nematode *Meloidogyne incognita*, adopt the following measures:

a. Use nematode free rooted cuttings for raising new plantations.
b. Apply talc based formulation of *Bacillus macerans* @ 10g/vine in basins (10^6 cfu/g) at the time of planting of vines or just before the monsoon period in established plantations.

Drench carbosulfan 25 EC 0.1% @ 50 ml / polybag (1.5 kg potting mixture) for the management of plant parasitic nematodes in black pepper nurseries. Recommended dose of carbosulfan is compatible with *Trichoderma harzianum* and *Pochonia chlamydosporia.*

Pochonia chlamydosporia (MTCC 5412) bioformulation, a native nematode ant agonistic fungus is recommended for field management of root knot nematodes (*Meloidogyne* sp.) and burrowing nematodes (to a lesser extent) infesting black pepper plants. The mode of application is 10 g formulation mixed with 2 kg of well decomposed farmyard manure or compost and applied to basin of the vine in the field. *Pochonia* is compatible with copper oxychloride and *Trichoderma harzianum* and does not have any non-target effects on other living beings.

Diseases

Phytophthora (Foot rot)

For controlling the disease, adopt the following management practices:

Phytosanitation

All infected or dead vines along the root system are to be removed and burnt. Wherever water stagnation is a problem, effective drainage of both surface and sub-soil is to be ensured. To avoid soil splash and consequent disease initiation and spread, a legume cover in the plantation should be ensured. Runner shoots are to be pruned or tied back to vines before the onset of monsoon. At the onset of monsoon, the branches of support trees may be lopped off to allow penetration of sunlight and avoid build up of humidity.

Apply 1 kg lime and 2 kg neem cake per standard per year as pre-monsoon dose. The application of neem cake should be four weeks after lime application.

Chemical control

For the control of *Phytophthora* foot rot, any of the following control measures can be adopted.

1. After the receipt of monsoon showers (May-June), all the vines are to be drenched over a radius 45-50 cm with copper oxychloride 2 g l^{-1} @ 5-10 litres per vine. This varies according to the age of the plant. A foliar spray with 1 per cent Bordeaux mixture is also to be given. Drenching and spraying are to be repeated just before the northeast monsoon. A third round of drenching may be given during October if the monsoon is prolonged.

2. After the receipt of a few monsoon showers (May-June), all the vines are to be drenched over a radius of 45-50 cm with (3 ml l^{-1}) potassium phosphonate @ 5-10 litres per vine. This varies according to the age of the plant. A foliar spray with 0.3 per cent potassium phosphonate is also to be given. A second drenching and spraying with 0.3 per cent potassium phosphonate is to be repeated just before the northeast monsoon. If the monsoon is prolonged, a third round of drenching may be given during October.

Biocontrol

Inoculate pepper vines with native arbuscular mycorrhizal fungi, *Trichoderma* and *Pseudomonas fluorescens* at the time of planting in the nursery and field and apply during the pre-monsoon period in the established plantations to control foot rot. In the field, apply the biocontrol agents around the base of the vine (see the chapter on biocontrol agents against plant pathogens).

Note: 1. All chemical control measures are prophylactic in nature and application of chemicals in advanced stages of disease will not be effective in combating the disease.

2. In *Phytophthora* sick fields, use only chemical control measures.

Replanting/rejuvenation

Total replanting has to be undertaken in gardens where the mortality is 50-60 per cent or above. Where the mortality is below 50 per cent, timely plant protection measures as described above should be given to all the existing vines as prophylactic measure and gaps filled up. Gap filling or replanting should be taken up only after a period of one year. At the time of replanting, soil drenching with copper oxychloride should be given. While replanting, farmers should be encouraged to use recommended varieties.

Fungal pollu (Anthracnose)

For the control of fungal pollu or anthracnose caused by *Colletotrichum gloeosporioides*, spray 1 per cent Bordeaux mixture, once before flowering starts (late June and early July) and then at berry formation stage (late August). Minimize shade in the garden.

Treat the cuttings before planting by immersing in a solution of carbendazim + mancozeb (1 g l^{-1}) for 30 minutes and spray Bordeaux mixture (1%) alternating with carbendazim (1 g l^{-1}) for effective control of *Colletotrichum* infection in nurseries.

Foliar spray of carbendazim @ 1g l^{-1} or a formulation containing combination of carbendazim + mancozeb @ 1g l^{-1} during the month of June can effectively control the disease.

Wherever *Phytophthora* foot rot management is undertaken properly, separate control measures for pollu disease may not be necessary.

Note: Since Bordeaux mixture application for pepper is to be given mostly at a time when the monsoon is very active, it is to be ensured that a sticker is added to the fungicide. The cheapest and most effective sticker is rosin washing soda mixture.

Rotting disease

For control of rotting disease of cuttings in the nursery, VAM and *Trichoderma* can be applied in the potting mixture. VAM inoculum consisting of infected root bits and soils can be applied @100 cc per kg of potting mixture and *Trichoderma* @ 1g kg^{-1} of potting mixture. For the control of foliar infection apply potassium phosphonate @ 3 ml $litre^{-1}$ at fortnightly interval. In case, biocontrol agents are not incorporated in the potting mixture, 1 per cent Bordeaux mixture spray at weekly interval may be resorted to. When the cuttings start germination, ensure good aeration in the nursery. Heavy watering, which causes water stagnation is to be avoided. Instead, light and frequent watering should be resorted to. Remove shade as soon as continuous rain sets in.

Phyllody

In certain pockets, instead of normal spike with berries, leaf-like structures are produced. This is caused by Phytoplasma. Such vines, if noticed, must be uprooted and destroyed. Planting material should not be collected from such vines.

Stunted disease

The symptoms due to this disease include shortening of internode and narrowing of leaves with mottling. Such leaves also become leathery and deformed. This is caused by a virus. Since the disease is systemic and

transmitted through planting materials, avoid collecting planting materials from such vines. Once it is noticed, uproot the vines to avoid further spread.

Waiting period of insecticide / fungicide

Dimethoate	20 days
Quinalphos	20 days
Mancozeb	30 days

Harvesting and processing

Black pepper

Black pepper of commerce is produced from whole, unripe but fully mature berries. The harvested berries are piled up in a heap to initiate browning. Then berries are detached from the stalk by threshing. Then they are spread on suitable drying floor. During sun-drying, berries are raked to ensure uniform drying and to avoid mould development. Drying the berries for 3-5 days reduces the moisture content to 10-12 per cent.The dried berries are cleaned, graded and packed in double lined gunny bags.

Blanching the berries in boiling water for one minute prior to sun drying accelerates browning process as well as the rate of drying. It also gives a uniform lustrous black colour to the finished product and prevents mouldiness of berries.

White pepper

White pepper is prepared from ripe berries or by decorticating black pepper. Bright red berries, after harvest are detached from the stalk and packed in gunny bags. The bags are allowed to soak in slow running water for about one week during which bacterial rotting occurs and pericarp gets loosened. Then the berries are trampled under feet to remove any adhering pericarp, washed in water and then sun dried to reduce the moisture content to 10-12 per cent and to achieve a cream or white colour. White pepper is garbled, sorted and packed in gunny bags. Approximately 25 kg white pepper is obtained from 100 kg ripe berries.

HACCP protocol for pepper products

The product specific hazard analysis and critical control points (HACCP) worksheet for different spice products *viz*., garbled pepper, white pepper, dehydrated green pepper (DGP) and pepper in brine are shown in Table 27 (A).

Table 27(A) Critical control points at various steps of pepper processing line

	CCP1	CCP2	CCP3	CCP4
Product	All 4 products*	All 4 products*	All 4 products *	Pepper in brine
Process stage Hazard	Reception Mycotoxin	Reception Chemical residue	Metal detection Metal	Brining Chemical, Microbiological
Monitoring procedure	Vendor/farmer selection, sample testing & analysis		On-line inspection	
Prerequisite programme	proper handling and storage	GAP	-	GMP, SSOP
Post occurrence control	No effective technique	-	Metal detector	No effective technique
Corrective action	Avoid entry to the process line		Hold and review	Water quality testing

**Common for garbled pepper, white pepper, DGP & pepper in brine*

Improved CFTRI method

Fully mature but unripe berries are harvested and boiled in water for 10-15 minutes to soften the pericarp. After cooling, the skin is rubbed off either mechanically or manually, washed and sun dried to obtain white pepper. Since no retting operation is involved, the product will be free from any unpleasant odour. However, white pepper produced by this method gives pepper powder of light brown colour due to gelatinisation of starch in contrast to pure white powder obtained by traditional method.

Decorticated black pepper

This is a form of white pepper produced by mechanical decortication of the outer skin of black pepper. This is generally done when white pepper is in short supply. The appearance of decorticated kernel is inferior to traditionally prepared white pepper, but is satisfactory when ground. Also the milling operation requires considerable skill to avoid excessive volatile oil loss.

Dehydrated green pepper

In this method, under-mature berries are harvested and subjected to heat treatment for inactivating the enzymes responsible for browning reaction.Then the berries are dehydrated under controlled conditions wherein maximum retention of green colour is obtained. Dehydrated green pepper after reconstitution in water resembles freshly harvested green pepper. The advantage is that the season of availability can be extended and the berries could be stored for a year or more. Dry recovery comes to 20 per cent.

Canned green pepper

Green pepper after harvest is preserved in two per cent brine solution and the product is heat sterilized. This product has the additional advantage over dehydrated green pepper in that it retains the natural colour, texture and flavour.

Bottled green pepper

Green pepper is preserved without spoilage in 20 per cent brine solution containing 100 ppm SO_2 and 0.2 per cent citric acid. Addition of citric acid prevents blackening of berries.

Cured green pepper

To overcome the disadvantages of poor texture and weak flavour of dehydrated greenpepper and the high unit weight and packing cost of canned and bottled green pepper, cured green pepper has been developed. Berries are thoroughly cleaned in water, steeped in saturated brine solution for 2-3 months, drained and packed in suitable flexible polyethylene pouches.

Freeze-dried green pepper

Most of the moisture from fresh tender green pepper is removed by freezing the berries at -30°C to - 40°C under high vacuum. The colour, aroma and texture of freeze-dried green pepper are superior to sun dried or mechanically dehydrated green pepper. Freeze-dried green pepper has 2 - 4 per cent moisture and is very light.

Pepper oil

Black pepper is crushed to coarse powder and steam distilled to obtain 2.5 to 3.5 per cent colourless to pale green essential oil which becomes viscous on ageing. It is used in perfumery and in flavouring. The drying percentage & oil content of KAU Varieties of Pepper are given as Table 28. Oil can also be distilled from white pepper but high price of white pepper and low oil yield do not favour its commercial production.

Pepper oleoresin

Extraction of black pepper with organic solvents like acetone, ethanol or dichloro-ethane

Table 28. Drying percentage and oil content of Panniyur varieties of pepper

Properties	Variety						
	P 1	P 2	P 3	P 4	P 5	P 6	P 7
Drying %	35.3	35.7	27.8	34.7	35.7	32.9	33.6
Piperine %	5.30	6.50	4.80	4.40	5.30	4.94	5.57
Oleoresin %	11.78	12.20	10.40	11.30	12.33	8.27	10.61
Essential oil %	3.31	3.40	3.12	3.12	3.80	1.33	1.50

provides 10-13 per cent oleoresin possessing the odour, flavour and pungent principles of the spice. The content of the pungent alkaloid piperine ranges from 4 to 6 per cent in dry pepper and 35 to 50 per cent in oleoresin. When freshly made, pepper oleoresin is a dark green, viscous, heavy liquid with a strong aroma. One kg of oleoresin when dispersed on an inert base can replace 15 to 20 kg of spice for flavouring purpose.

TURMERIC (*Curcuma longa*)

Turmeric is a tropical herb and can be grown on different types of soil under irrigated and rainfed conditions. Rich loamy soils having good drainage are ideal for the crop. It is a shade tolerant crop with shallow roots suitable for intercropping and also as a component crop in the homesteads where low to medium shade is available.

Preparation of land

Prepare the land to a fine tilth during February-March. On receipt of pre-monsoon showers in April, prepare beds of size 3 m x 1.2 m with a spacing of 40 cm between beds.

Seed material

Whole or split mother rhizomes are used for planting. Select well developed, healthy and disease free rhizomes. Treat the rhizomes in any of the copper oxychloride fungicides and store in cool, dry place or in earthen pits plastered with mud and cowdung.

Varieties

Cultivars: Tekurpetta, Sugantham, Kodur, Armoor, Alleppey.

Improved varieties: Suvarna, Suguna, Sudarshana, Prabha, Prathibha, Kanthi, Sobha, Sona, Varna, IISR Kedaram and IISR Alleppey Supreme (Resistant to leaf blotch).

Season and method of planting

Plant during April with the receipt of pre-monsoon showers. Take small pits in the beds in rows with a spacing of 25cm x 25cm. Plant finger rhizomes flat with buds facing upwards and cover with soil or dry powdered cattle manure. The seed rate is about 2000-2500 kg ha^{-1}.

Manuring

Apply cattle manure or compost as basal dose at 40 t ha^{-1} at the time of land preparation or by spreading over the beds after planting. Apply $N:P_2O_5:K_2O$ @ 30:30:60 kg ha^{-1}. Full